

EDUCATIONAL PSYCHOLOGY

(Text Book Based on the New Syllabus introduced in 2021 by
the Tamilnadu Teacher Education University, Chennai)

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PREFACE

This text book titled "EDUCATIONAL PSYCHOLOGY" has been written, based on the new syllabus for the I Semester paper of the B.Ed. Degree Course of the Tamil Nadu Teacher Education University. The five units of this subject has been discussed under the following headings.

1. Educational Psychology and Human Growth Development,
2. Attention, Perception and Memory, 3. Motivation and Learning, 4. Intelligence and Creativity, 5. Personality.

To facilitate the easy understanding of the subject by the teacher-trainees, the contents have been presented under different headings and subheadings. As of our previous works, this book is also written in a lucid style with vivid explanations and apt examples. To help students prepare for their examinations, review questions and the expected answers for the same have been provided at the end of each lesson. A set of Model Question Papers based on the University examination pattern with the expected answers for the same cited in the book, have been provided at the end of this book to help students who are to take their B.Ed. Degree Examination under the new syllabus for the first time.

I am thankful to, **Mr.C.R. Manivasagan**, M.Com.,M.Ed, former Office Superintendent of Meston College of Education, Chennai - 14, for his pains-taking efforts in proof reading this book and **Mr. C. Boopalan** of **Printech India** for neatly printing this book.

Constructive suggestions are quite welcome to refine this book further. I earnestly solicit the continued support of teacher-educators and student-teachers, as they have been offering for my previous works. It will be highly satisfying to me if this book proves useful in the hands of student community.

K. Nagarajan

Chennai (Dr.K. Nagarajan)

10.12.2021

1. Symbols (A), (B), (C) used in Review Questions refer to Essay, Short Answer and Very Short Answer type questions respectively.
2. Essay type questions are to be answered in about 5 pages.
3. Short Answer type questions are to be answered in about 2½ page.

CONTENTS

Unit	Lesson	Topic	Page
I		EDUCATIONAL PSYCHOLOGY AND HUMAN GROWTH AND DEVELOPMENT	1
	1:00	Introduction	1
	1:01	Definition of 'Psychology'	1
	1:02	Historical Development of Psychology and its Emergence as a Separate Discipline	3
	1:03	Definition of Educational Psychology	10
	1:04	Nature of Educational Psychology	10
	1:05	Scope of Educational Psychology	11
	1:06	Differences Between General and Educational Psychology	15
	1:07	Significance of Educational Psychology	16
	1:08	Meaning of the term 'Growth'	20
	1:08:1	Characteristics of 'Growth'	20
	1:09	Meaning of the term 'Development'	21
	1:09:1	Characteristics of 'Development'	21
	1:10	Comparison between 'Growth' and 'Development'	23
	1:11	Meaning of Maturation	24
	1:12	Relationship between Growth, Development and Maturation	24
	1:13	Dimensions of Human Development	25
	1:14	Developmental Phases	28
	1:15	Important Phases of Development	28
	1:16	Physical Development from Infancy to Adolescence	30
	1:17	Intellectual Development from Infancy to Adolescence	32

CONTENTS

Unit	Lesson	Topic	Page
1:18		Social Development from Infancy to Adolescence	33
1:19		Emotional Development from Infancy to Adolescence	36
1:20		Stages of Moral Development in Children	38
1:20:1		Jean Piaget's Stages	38
1:20:2		Kohlberg's Stages	39
1:21		Period of Adolescence	40
1:22		Special Characteristics of Adolescents	41
1:23		Problems of Adolescents	44
1:23:1		Identity Crisis	44
1:23:2		Generation Gap	45
1:23:3		Negative and Non-Conformist Attitude	45
1:23:4		Domination of Sexual Urge	46
1:23:5		Hesitation and Confusion in Selecting a Vocation	46
1:23:6		Emotional Swings	46
1:23:7		Aspiring for Autonomy	46
1:23:8		Peer Group Dominance	47
1:24		Helping the Adolescents to Overcome their Problems	47
1:25		Developmental Tasks	51
1:25:1		Uses of the Knowledge of Developmental Tasks	53
1:25:2		Harms due to Insistence of Developmental Tasks	54
1:26		Developmental Tasks for the Various Stages of Development	55
1:27		Education and Developmental Tasks	57
		Conclusion	58
		Review Questions	59

CONTENTS

Unit	Lesson	Topic	Page
2:00		ATTENTION, PERCEPTION AND MEMORY	63
2:00		Introduction	63
2:01		Meaning of Attention	63
2:01:1		Definition of Attention	64
2:02		Types of Attention	64
2:03		Nature / Characteristics of Attention	66
2:04		Determinants of Attention	67
2:04:1		External Factors of Attention	68
2:04:2		Internal Factors of Attention	72
2:05		Ways and Means of Securing Better Attention in the Class	73
2:06		Sensory Organs and Sensations	75
2:06:1		Sensation	75
2:07		Perception	76
2:08		Determinants of Perception	76
2:09		Laws of Perceptual organisation	77
2:09:1		Law of Pragnanz	77
2:09:2		Law of Proximity	78
2:09:3		Law of Similarity	78
2:09:4		Law of Closure	79
2:09:5		Law of Good Continuation	79
2:10		Errors in Perception	80
2:10:1		Illusion	80
2:10:2		Hallucination	82
2:11		Meaning of 'Memory'	82
2:12		Kinds of Memory	83
2:12:1		Short and Long Term Memories	84
2:13		Factors Affecting Memory	86

CONTENTS

Unit	Lesson	Topic	Page
2:14		Signs of Good Memory	87
2:15		Techniques of Promoting Better 'Memory'	88
		Conclusion	90
		Review Questions	90
3:00		Motivation and Learning	92
3:00		Introduction	92
3:01		Meaning and Definition of Motivation	92
3:02		Definition of Motives	93
3:03		Characteristics of Motives	94
3:04		Kinds of Motives	95
3:05		Theories of Motivation	96
3:05:1		Maslow's Theory of Hierarchical Needs	96
3:05:1:01		Educational Implications of Maslow's Theory	99
3:05:1:02		Differences Between Deficiency and Growth Needs	100
3:05:1:03		Characteristics of Self-actualisers	101
3:06		Concept of 'Level of Aspiration'	102
3:07		Two Types of Motivation	103
3:07:1		Extrinsic Motives	103
3:07:2		Intrinsic Motives	104
3:07:3		Merits and Limitations of Extrinsic and Intrinsic Motives	104
3:08		Motivational Strategies in the Classroom	104
3:09		Meaning and Definition of 'Learning'	107
3:10		Nature of Learning	108
3:11		Factors Influencing Learning	109
3:12		Learning Theories	111

CONTENTS

Unit	Lesson	Topic	Page
3:12:1		Two Major Systems of Learning Theories	112
3:12:2		Differences between Association and Field Theories	112
3:13		Thorndike's Connectionism	115
3:13:1		Characteristics of Trial and Error Learning	116
3:13:2		Thorndike's Puzzle Box Experiment	116
3:13:3		Thorndike's Laws of Learning	117
3:13:4		Limitations of Thorndike's Theory of Learning	119
3:14		Classical Conditioning	120
3:14:1		Pavlov's Experiment	121
3:14:2		Educational Implications of Conditioning Theory of Learning	123
3:14:3		Limitations of Conditioning Theory of Learning	124
3:14:4		Laws of Conditioning	124
3:15		Concept of 'Reinforcement'	126
3:16		Positive and Negative Reinforcers	126
3:17		Skinner's Operant Conditioning	127
3:17:1		Skinner's Experiments	128
3:17:2		Educational Implications of Skinner's Theory	129
3:18		Comparison Between Classical and Operant Conditioning	130
3:19		Differences Between 'Reinforcement' and 'Feedback'	131
3:20		Jean Piaget's Theory of Cognitive Development	131
3:20:1		Sensory - Motor Stage	132
3:20:2		Pre-operational Stage	134
3:20:3		Concrete Operational Stage	135

CONTENTS

Unit	Lesson	Topic	Page
	3:20:4	Stages of Formal Operational Stage	136
	3:21	Educational Implications of Piaget's Theory	139
	3:22	John Dewey's Theory of Constructivism	140
	3:22:1	Meaning of Knowledge Construction / Constructivism	140
	3:22:2	Nature of Constructivist Learning	141
	3:22:3	Role and Functions of Teacher in the Constructivist Classroom	143
	3:22:3:01	Making Decisions Before the Lesson Begins	145
	3:22:3:02	Setting the Learning Tasks for the Lesson	145
	3:22:3:03	Monitoring and Intervening During the Group Work	145
	3:22:3:04	Evaluating the Product and Process of Group work	146
	3:23	Humanistic Psychology and its Basic Premises	146
	3:24	Humanistic Psychologists' View of 'Learning'	147
	3:25	Carl Roger's Theory of Fully Functioning Person	149
	3:25:1	Characteristics of Fully Functioning Person	149
	3:26	Educational Implications of Learning Theories	151
		Conclusion	152
		Review Questions	153
4		INTELLIGENCE AND CREATIVITY	156
	4:00	Introduction	156
	4:01	Meaning of Intelligence	156
	4:02	Definitions of Intelligence	157
	4:03	Characteristic of an Intelligent Person	158

CONTENTS

Unit	Lesson	Topic	Page
	4:04	Types of Intelligence	159
	4:05	Theories of Intelligence	160
	4:05:1	Unitary Theory or Monarchic Theory	160
	4:05:2	Spearman's Two-factor Theory	161
	4:05:3	Thorndike's Multifactor Theory	162
	4:05:4	Thurstone's Group-factor Theory	163
	4:05:4:01	Contributions of Thurstone's Theory of Intelligence	165
	4:05:5	Guilford's Structure of Intellect	165
	4:05:5:01	Educational Implications of Guilford's Structure of Intellect Model	169
	4:06	Howard Gardner's Theory of Multiple Intelligence	169
	4:06:1	Gardner's Eight Types of Intelligence	170
	4:07	Measurement of Intelligence	172
	4:07:1	Difficulty in Measuring Intelligence	172
	4:07:2	Some of the Standardised Intelligence Tests	173
	4:07:2:01	Stanford-Binet Test of Intelligence	173
	4:07:2:02	Concept of Mental Age	174
	4:07:3	Concept of Intelligence Quotient (I.Q.)	174
	4:07:3:01	Wechsler's Adult Intelligence Scale (WAIS)	175
	4:07:4	Classroom Teacher Assessing the Intelligence of his Students	176
	4:08	Types of Intelligence Tests	179
	4:08:1	Individual and Group Tests	179
	4:08:1:01	Advantages of Group Tests of Intelligence	181
	4:08:1:02	Limitations of Group Tests of Intelligence	181
	4:08:2	Verbal and Performance Tests	182

CONTENTS

Unit	Lesson	Topic	Page
	4:08:2:01	When to Use Performance Tests?	183
	4:08:2:02	Limitations of Performance Tests	183
	4:09	Uses of Intelligence Tests	184
	4:10	Limitations of Intelligence Tests	185
	4:11	Culture-Fair Tests	186
	4:12	Meaning of 'Creativity'	186
	4:12:1	Nature (Characteristics) of Creativity	188
	4:13	Stages in the Process of Creative Thinking	189
	4:14	Differences Between Intelligence and Creativity	191
	4:15	Identification of the Creative Children	192
	4:15:1	Use of Tests of Creativity	192
	4:15:2	Use of Non-testing Techniques	193
	4:16	Fostering Creativity among Children	194
		Conclusion	198
		Review Questions	198
5:00		PERSONALITY	201
5:00		Introduction	201
5:01		Meaning of the term 'Personality'	201
5:02		Definitions of Personality	202
5:03		Characteristics of Personality	203
5:04		Uses of the Knowledge of Personality to Teachers	203
5:05		Factors Influencing Personality Development	204
5:05:1		Biological Factors	204
5:05:2		Social Factors	206
5:05:3		Psychological Factors	208

CONTENTS

Unit	Lesson	Topic	Page
	5:06	Theories of Personality	209
	5:06:1	Type Approach to Personality	209
	5:06:1:01	Galen's Classification	209
	5:06:1:02	Kretschmer's Classification	210
	5:06:1:03	Spranger's Classification	210
	5:06:1:04	Sheldon's Classification	211
	5:06:1:05	Lippitt's Classification	212
	5:06:1:06	Ogburn's Classification	213
	5:06:1:07	Jung's Classification	213
	5:06:2	Limitations of the Type-Approach	213
	5:07	Trait-Approach to Personality	214
	5:07:1	Meaning of the term 'Trait'	214
	5:07:2	Cattell's Trait Approach to Personality	214
	5:07:3	Eysenck's Type cum Trait Approach to Personality	215
	5:07:3:01	Characteristics of 'Extroverts' and 'Introverts'	216
	5:07:4	Mertis and Limitations of Trait Approach	216
	5:07:4:01	Usefulness of Trait Approach	216
	5:07:4:02	Limitations of Trait Approach	217
	5:08	Freud's Psycho-analytic Theory of Personality	217
	5:08:1	Freudian Stages of Personality Development	220
	5:09	Integrated Personality	221
	5:10	Personality Assessment	222
	5:11	Approaches and Methods of Assessing Personality	223
	5:11:1	Interview Method	224
	5:11:1:01	Limitations of Interview Method	225

CONTENTS

Unit	Lesson	Topic	Page
	5:11:1:02	Uses of Interview	225
	5:11:2	Rating Scales	226
	5:11:3	Questionnaire	228
	5:11:3:01	Model Questions in a Questionnaire	228
	5:11:3:02	Limitations of Questionnaire	229
	5:11:3:03	Merits of Questionnaire	230
	5:11:4	Projective Techniques	230
	5:11:4:01	Salient Features of Projective Tests	231
	5:11:4:02	Rorschach Ink Blot Test	232
	5:11:4:03	Murray Morgan's Thematic Apperception Test (T.A.T)	234
	5:11:4:04	Situational Tests	234
	5:11:4:05	Free Association Tests	236
		Conclusion	236
		Review Questions	237

Unit – I

EDUCATIONAL PSYCHOLOGY AND HUMAN GROWTH AND DEVELOPMENT

1:00 Introduction

In this Unit, Meaning and Definition of Psychology, Historical Development of Psychology and its emergence as a separate discipline, Definition of Educational Psychology, its scope and importance, Meaning and definition of the terms, 'Growth' and 'Development', their characteristics, Comparison between Growth and Development, Meaning of Maturation and its relationship with growth and development, Dimensions of human development, Developmental Phases of Human Development, Physical and Intellectual Development from Infancy to Adolescence, Special characteristics of Adolescence, Problems of Adolescence, Developmental Tasks and Developmental Tasks at various stages of Development are to be discussed at length.

1:01 Definition of 'Psychology'

Etymologically, the word "Psychology" derives its meaning from the two Greek words 'psyche' which means 'soul' and 'logus' which means 'science' or 'knowledge'. Thus 'Psychology' literally means the science of the soul. This definition of 'Psychology' has

been given up long ago. Today Psychology is defined as the science of behaviour, the way human beings behave and the reasons for their behaviour. The term behaviour should be understood in a broad manner, to refer not only to observable overt bodily activities (which alone should be the subject matter of psychological investigations as per behaviourism) but also covert and inferred behaviour which could be designated as mental or conscious activity. As such psychology is now defined as **"a science that systematically studies and attempts to explain observable behaviour and its relationship to unseen 'mental' processes (experiences) that go on inside the organism and to external events in the environment"**. The goals of psychology are description, explanation, prediction and control of behaviour including subjective experiences.

By observing events systematically, psychologists can predict in advance, things that are to follow. For example if a student gets fail marks in most of the monthly tests in mathematics, it could be predicted that he will fail in maths in the annual examinations. The reasons for poor performance in maths (lack of interest in maths, lack of motivation, inadequate efforts, negative attitude towards maths etc.) could also be explained. A maths teacher with sound knowledge of psychology could plan for necessary behaviour changes in students so as to prevent them scoring poor marks in maths.

1:02 Historical Development of Psychology and its Emergence as a Separate Discipline

Till 19th century, psychology was studied only as a branch of philosophy. When we look at the original meaning of the word psychology this will become clear. The word psychology comes from the two Greek words, 'psyche' (soul) and 'logus' (science); so the root meaning of the word psychology is that it is a science of soul. This is the traditional approach to the study of psychology. Traditional psychologists attempted to study the location of the soul, its nature and its state after the death of the individual. When we are not sure about the location of the soul, attempting to investigate its nature is akin to the efforts of a blind man, searching for a black cat and that too in a place where it is not. So the definition of psychology as the study of soul was given up.

Then it was defined as a science of mind. Mind functions in three domains viz. Cognitive (thinking), Affective (feeling) and Conative (willing). A mental act is different from a physical act. If a piece of iron is placed near a magnet, the iron moves towards the magnet. This is a physical act and no thinking, feeling or willing is involved on the part of iron or magnet. But on the contrary, when we place a dish of milk before a cat, the cat moves towards the milk and drinks it. This is a mental act because the cat driven by hunger motive, knowing (thinking) the white liquid is milk, which it likes (feeling part) to drink, it actually moves (willing part) towards the

milk. Thus every human act or behaviour emanates from an internal **motive**, as the British psychologist **McDougall** puts it.

According to **William McDougall**, who propounded the '**Hormic School of Psychology**' (Purposivism), every response of man is due to a purpose or inner motive called '**Instinct**'. McDougall criticised behaviourism. He stated that response occurs, not always due to the occurrence of a stimulus. It is not necessary that we feel the desire to eat when we look at sweets. Desire to eat depends upon the hunger motive. Different motives result in different responses. According to McDougall, it is the instinct that motivates human behaviour. He further believes that an emotion is present in any instinctive activity. Each instinct associated with some emotions (created by subjective experience one had from his environment) otherwise known as '**sentiments**' become the centre of all activities. Without them no activity is possible. Though all of us have similar instincts, they get modified according to one's environment. Our behaviour depends upon the modification of the instincts or our sentiments. In other words, sentiments are the motives of our behaviour and these sentiments may be analysed into instincts and emotions. Thus instincts and emotions are the bases of human behaviour according to Hormic Psychology.

Titchner, who developed the theory of '**Structuralism**' believed that mind is the basis of all human actions. According to him there are three states of mind - cognitive,

affective and conative. The mind was regarded as consciousness and the ultimate elements of it are sensation, feelings and images. The nature and structure of consciousness can be studied through **introspection**. To introspect means to look within. **Psychology** thus came to be defined as a science of consciousness.

Behaviour psychologists like **J.B. Watson** of U.S.A and others questioned the validity of introspection and said that psychology should be an objective science. They bitterly criticised introspection as unreliable. They defined psychology as a science of behaviour. They tried to define behaviour in terms of stimulus and response, popularly known as S-R formula. For every stimulus, there is a response. The response to stimulus is behaviour. If you are pricked with a pin, there is a jerk of the body. 'Pricking' is the stimulus and the 'jerk' is the response or behaviour. A simple behaviour consists of the basic unit of behaviour i.e. S-R. Any complex behaviour can be analysed in terms of S-R Units chained sequentially. For behaviourists, environment is the determinant of one's response or behaviour and heredity has no influence what so ever. They also deny **McDougall's Theory of Instincts**. **Bernard**, the behaviourist states that we do not allow our instincts like fear, sex, acquisition, affiliation etc. to operate as they are; but we modify them in the midst of our social environment. Behaviourists also deny any role for mind in determining our behaviour. They argue that we talk because of the movement of our tongue and not because of thinking or functioning of the mind.

As a revolt against behaviourism and structuralism **Gestalt School of Psychology** emerged in 1912. Gestalists advocated that everything should be viewed as a whole and studied in its totality. The researches of **Werthemer, Kohler and Koffka** contributed a lot to the development of this school. The German word '**Gestalt**' refers to '**configuration**', '**whole**', or '**totality**'. We generally perceive the whole and not its parts.

The 'whole' is not the addition of the parts but something more than that. When we purchase a bicycle, we only perceive it as a whole cycle and not as spokes, wheel, saddle, chain, mudguard etc. Sometimes we fail to notice even some missing parts like bell or lock. So according to Gestalists **perception is unified and learning is nothing but reorganisation of the field of perception**. Their main areas of investigation are perception and learning. To perceive an object or event, apart from our sensation of the object or event, our previous experiences about them are also called into play. Things are perceived in the canvas of our background experience.

Gestalists advocate that human behaviour is based on one's perception and perception is influenced by our past experiences stored in our mind. For them behaviour is physical (as behaviourists believe) as well as mental or cognitive. A particular stimulus does not elicit the same response from every one. People express different responses as per their previous experiences. On seeing a snake some may flee due to fear but few others may be brave enough to catch it.

Then came the Viennese physician **Sigmund Frued** (1856-1939) on the scene with a bang proclaiming the school of psychology called '**Psycho-analysis**'. While structuralism gave emphasis for the conscious mind, psycho - analysis emphasized the role of unconscious mind in the actions of individuals. Since psycho-analysis goes deep into the subteranean level of the mind, this is also known as **depth psychology**. The other schools of psychology deal only with surface activities. But psycho-analysis delves deep into the roots and springs of human behaviour. The unconscious mind is the seat of repressed tendencies and desires, not approved by the society. Sex plays an important role. According to Frued in the drama of human existence, sex is the villain of the peace. Frued holds that early childhood experiences determine the development of personality. He speaks of human personality made up of three major systems - **Id** (operating at unconscious level), **Ego** (operating at conscious and sub-conscious levels) and **Super Ego** (operating at conscious level).

Frued's disciples **Adler and Carl Jung** broke away from him and founded their own schools of psychology viz., '**Individual Psychology**' and '**Analytical Psychology**' respectively. Unlike Frued, Jung stresses past experiences of the individual. He thinks religion has a lot of therapeutic value. According to Adler, birth order of an individual in the family has much influence in determining one's way of life.

Piagets's Cognitive psychology, Humanistic

psychology of **Carl Rogers, Maslow** and others have also significantly contributed to the growth of psychology and made it attain today the status of positive behavioural science. As a pure science, psychology is concerned with systematic study of behaviour and verification through experimentation. We can bring psychology under the category of 'Bio-social science'. The components of behaviour are :

- i) The conscious experiences of which the organism is aware of, like that of being hungry or having pain when injured, etc. and
- ii) The unconscious process: (eg.) without any specific reason we get irritated with some people, become friendly with others etc.

To attain this significant status, many had contributed to the growth of psychology, of whom the following are notable.

- i) **E.H. Weber** : Forerunner for conducting psychology experiments; examined the methods of measuring sensory experiences based on which formed the Weber's Law.
- ii) **G.T. Fechner (1860)** : Published his book Elements of Psycho-physics which investigated problems in sensation and perception
- iii) **Wilhelm Wundt (1879)** : Established the first psychology laboratory at 'Leipzig' in Germany. He investigated on attention, reaction time, memory association, emotions etc. He

established the method of introspection as a technique of data collection. He is the founder of structuralist school of psychology.

- iv) **Sir Francis Galton (1822-1911)** : Studied individual differences using statistical techniques (use of correlation coefficients).
- v) **Ivan Pavlov (1849-1936)** : Studied conditioned reflexes.
- vi) **E.L. Thorndike (1874-1947)** : Formed the Trial and Error theory of learning and laws of learning; advocated the Multifactor theory of intelligence.
- vii) **J.M. Cattell (1860-1944)** : Developed quantitative methods in psychology; was an assistant to Wundt, established his psychology laboratory at Columbia in America.
- viii) **E.B. Titchner (1867-1929)** : Represented structural school in America
- ix) **G.Stanley Hall (1846-1924)** : Wundt's famous American student; began systematic study in child psychology in the U.S.
- x) **Mesmer** : Clinical-psychological methods.
- xi) **Jean Piaget** : Theory of cognitive development in children.
- xii) **Sigmund Freud (1856-1939)** : Founding father of psycho-analysis and known as father of Modern Psychology.

- xiii) **Alfred Binnet (1905)** : Developed the concept of mental age; constructed the first intelligence test.
- xiv) **B.F. Skinner** : Operant conditioning in learning.
- xv) **Weschler** : Developed Adult Intelligence Scale.

1:03 Definition of Educational Psychology

Educational psychology is defined as that branch of psychology which is concerned with psychological researches as applied to any or all aspects of educational processes and practices. It is in relation to problems like learning, teaching and training. Classroom is the laboratory for the educational psychologists. By applying the principles and laws of psychology in educational situations, educational psychology tries to solve the different problems faced in the educative process so as to make it more effective and efficient (says **Kolesnik**, the Russian Psychologist).

Psychology is the science of behaviour. Education is the deliberate process of modifying one's behaviour (knowledge, skills and attitude) through a sequence of systematically planned experiences to achieve the predetermined goals and objectives (**Redden**). Therefore educational psychology could be considered as the science of modifying the learner's behaviour so as to refine his personality and make him an efficient and responsible citizen.

1:04 Nature of Educational Psychology

Educational psychology has made tremendous

advancement and gradually established itself as an independent study. Most of the educational problems come under its purview. Child education, diagnostic and remedial education, education for the gifted and handicapped, factors affecting learning, principles of evaluation, etc. bear proof to the claim that educational psychology is a special field of study. The study of classroom climate, techniques of behaviour modification, programmed learning, micro-teaching, software and hardware approaches, computer assisted instruction are all the areas developed by the educational psychologists aimed at improving the interaction between the teacher and the taught. Advanced techniques of instruction are through and through psychological by nature.

1:05 Scope of Educational Psychology

As **Blair, Jones and Simpson** say, educational philosophy is primarily concerned with the question of what should be done and educational psychology attempts to answer the question of how it can be done. Educational psychology for most part is interested in means rather than ends.

1. **Kolesnik** mentions the following problems dealt by educational psychologists as forming the scope of educational psychology :

- i) Differences among students
- ii) Motivation
- iii) Methods of instruction
- iv) Evaluation
- v) Classroom management

- vi) Mental health
- vii) Character formation

2. H.C. Lindgren opines that Educational Psychology revolves around the behaviour of the learner in educational situations and as such it should limit itself to the teaching - learning process and the educational environment. Of course the other elements involved in the educational process are the learning experiences provided and the teacher who organises them. Thus the basic elements of the content of Educational Psychology are:

- i) The Learner
- ii) The Learning Experiences
- iii) The Learning process
- iv) The Learning situation or environment
- v) The Teacher

(i) The Learner : The first focal area of Educational Psychology is the learner around whom the whole subject-matter is knitted. The subject-matter of this area acquaints us with the need of knowing the learner and deals with the techniques of knowing him well. The innate abilities and capacities of the individual, individual differences and their measurement, the overt, covert, conscious and unconscious behaviour of the learner, the characteristics of his growth and development at each stage beginning from childhood to adulthood, are studied in this section of Educational Psychology.

(ii) The Learning Experiences : Educational Psychology helps in deciding what learning experiences

are desirable, at what stage of growth and development of the learner so that these experiences could be acquired with great ease and satisfaction. In this area, Educational Psychology has the subject matter which deals with the knowledge and principles of psychology facilitating the selection of desirable learning experiences for the learner.

(iii) The Learning Process : After knowing the learner and deciding what learning experiences are to be provided, the emerging problem is to help the learner in acquiring these experiences with ease and convenience. Therefore around this pivot, Educational Psychology deals with the nature of learning and how it takes place and it contains the topics such as laws, principles and theories of learning, motivation, transfer of learning, remembering and forgetting, perceiving, concept formation, thinking and reasoning process, problem-solving etc.

(iv) The Learning Situation : This focal area of Educational Psychology deals with the environmental factors and learning situations which come midway between the learner and the teacher. Topics like classroom climate and group dynamics, techniques and aids which facilitate learning, learning situation, evaluation techniques and practices, guidance and counselling etc. which help in the smooth functioning of the teaching-learning process are studied in this section.

(v) The Teacher : The teacher himself is a very important element in the teaching-learning process.

Therefore the Psychology of being a teacher (his motivations, conflicts, effectiveness, professional growth, mental health etc.) also included in the scope of Educational Psychology so as to inspire him to become a successful teacher and play his role properly in the process of education.

3. **Garrison et. al.** recognise the following as major divisions of educational psychology :

1) *The child and his development* : The course of development, nature of intelligence, language and thinking, socialisation and its role, etc.

2) *Learning and educative process* : Learning and motivation, the learning of skills, knowledge, understanding and problem-solving, character development, etc.

3) *Evaluating Pupil-Growth* : Methods of pupil-evaluation, studying the individual child, learning difficulties at school, evaluating the results of instruction, etc.

4) *Guiding the child* : Personality intergration, adjustment problems of the child, pupil-teacher relations, etc.

In short, speaking with **E.A. Peal**, it might be said that **educational psychology broadly deals with the nature of learning, the growth of human personality, the differences between individuals and the study of the person in relation to society.**

1:06 Differences Between General and Educational Psychology

The basic difference between General Psychology and Educational Psychology appears to be the matter of focus. General psychology gives a comprehensive view of human behaviour. Educational psychology highlights the behaviour of the learner. **Learner, learning processes and learning situations are the three focal areas of educational psychology.**

The learner has the most important place in the classroom. One important factor that the teacher should know about the learner is that there are individual differences among learners. There are gifted, average and slow learners in the class. The class is generally not homogeneous but heterogeneous. The next area is learning process. An effective learning process is one where the instruction is individualised. The student's perception, thinking, reasoning, intelligence, remembering -all come under learning process. The third area is the learning situation, that includes the conditions like physical factors, the tone of the class, attitude and behaviour of the teacher, etc. In short, we call this as classroom climate. Generally there is diversity in the classroom and Educational Psychology aims to prepare the teacher to realize the individual differences among his pupils and plan accordingly to maximize the achievement of each individual learner entrusted to his care.

1:07 Significance of Educational Psychology

Educational psychology helps the teacher in the following ways :

A) Contribution to Theory of Education :

1) To understand the development characteristics of children : To be a successful doctor, one must have the required professional knowledge and skills as well as the knowledge about the nature of the patient whom he wants to treat. In the same way, if one wants to be a successful teacher, he must know about the science of behaviour of learners. Children pass through different stages of development as infancy, childhood and adolescence. These developmental stages have their own characteristics. If the prospective teacher knows these characteristics he can utilise them in imparting instruction and moulding their behaviour according to the specified goal of education.

2) To understand the nature of classroom learning : To instruct effectively in the class, the teacher must understand the principles of learning and various approaches to the learning process, problems of learning and their remedial measures. It also gives the knowledge of various approaches to understand the learning process, factors affecting it, and guidance for effective learning.

3) To understand individual differences and adjust his teaching to the needs and the requirements of the class which has a great range of individual differences.

4) To understand effective teaching methods : Classroom teaching is not dependent on any one theory. It is related and uses several theories of teaching-learning. The teacher must be acquainted with knowledge of various theories in order to organise his class-room teaching. Educational psychology provides us with the knowledge of different approaches evolved to tackle the problems of teaching at different age levels.

5) To understand the causes of the problems of children which occur at different age levels and successfully solve them.

6) Knowledge of mental health : Mental health of the teacher and the taught is very important for efficient learning. The teacher from the study of psychology can know the various factors which are responsible for the mental ill-health and maladjustment. He can prevent maladjustment in children provided he is equipped with the fundamental knowledge of mental hygiene.

7) Curriculum construction : Psychological principles are used in formulating curriculum for different stages. Needs of the pupils, their developmental characteristics, learning pattern and needs of the society, all these are to be incorporated in the curriculum.

8) Measurement of learning outcome : Psychological tools help the teacher to assess the learning outcome of the students. He can also evaluate his teaching methods and in the light of the performance of his students can modify his strategy of teaching.

- 9) Guidance for the education of exceptional children.
- 10) Helps to develop positive attitude.
- 11) Understanding of group dynamics.

B) Contribution to Practice of Education

1) Problem of discipline : Traditional teachers believed in the dictum "spare the rod and spoil the child". But modern teachers who have the knowledge of educational psychology realise that the use of corporal punishment is inhumane. Teachers now tackle the problem of indiscipline by examining the causal factors leading to them in a more scientific way. The teacher has changed his attitude from an autocrat to democrat.

2) Use of A.V. aids : Teaching with the use of aids that affect multiple senses of the learner is more effective and lasting. Thus the modern teachers resort to the use of hardware and software approaches in educational technology to make their teaching more concrete and stable.

3) School and classroom administration has been psychologised and made participative rather than dictatorial in nature.

4) Time table : Now subjects are incorporated in the time-table taking into consideration their difficulty level and fatigue index. No two difficult subjects are taught in successive periods.

5) Co-curricular activities : Teachers used to give undue importance to the theoretical subjects in schools. Activities like debate, drama, scouting and games were supposed to be a wastage of time. Now we give these activities due importance for the harmonious development of the personality of children.

6) Use of innovative methods : Several innovative ideas resulting from research studies in psychology have been introduced to improve the teaching-learning process. Activity-centred teaching, discussion method, micro-teaching, programmed instruction, and non-graded school classes at the primary stage are some of the important innovations.

7) Production of text books : We now write text books taking into consideration the intellectual development of children, their needs and their interests at different age levels.

Undoubtedly the study of educational psychology may be very helpful to equip our prospective teachers with necessary skills to deal with classroom teaching-learning problems. Summing up we can say in the words of W.B. Kolesnik, that **the purpose of educational psychology is not to give specific, definitive answers to questions about methods of teaching or techniques of dealing with children, not to arm the teacher in advance with pre-packaged solutions to the many practical problems that he is likely to encounter in the classroom. Its purpose, rather, is to**

introduce him to psychological theories and research findings about learners and the learning process, and to a method of thinking psychologically, that may help him answer his own questions, reach his own conclusions and solve his own problems.

1:08 Meaning of the term 'Growth'

In the strict sense of terminology the two terms growth and development have different meanings. The term 'growth' is used purely in the physical sense. It generally refers to increase in size, length, height and weight. Changes in the quantitative aspects, which could be objectively observed and measured, come into the domain of growth. Growth is one of the components of the developmental process. In a sense, development in its quantitative aspect is termed as growth.

1:08:1 Characteristics of 'Growth'

- 1) Growth refers to increase caused by becoming larger and heavier.
- 2) It is quantitative, additive, and augmental.
- 3) It is objectively observable and measurable.
- 4) Growth does not continue throughout life, it stops when maturity has been obtained.
- 5) Growth may or may not bring development. For example, a child may grow (in terms of weight) by becoming fat, but this growth may not bring any functional improvement (qualitative change) or development.

- 6) Rate of growth is not uniform. It proceeds more rapidly in the early years of life but slows down in the later years of infancy. Again, at puberty, there is sudden rise in the speed of growth.
- 7) There exist wide range of individual differences among children with respect to growth.
- 8) The rate of growth of different parts of the body is different.

1:09 Meaning of the term 'Development'

'Development' implies overall changes in shape, form or structure resulting in improved working or functioning. It indicates the changes in the quality or character rather than in quantitative aspects. Development is the result of growth, maturation and learning.

1:09:1 Characteristics of 'Development'

1. Development is progressive; series of changes occur in an orderly, predictable pattern as a result of maturation and experience. Development is 'directional' and 'sequential'.
2. Development is continuous in all areas of mental activity. It does not stop with puberty as in the case of growth.
3. The goal of development is to enable the individual to adapt to the environment in which he lives, i.e. development results in improved functioning of the individual.

4. Development is a complex process of integrating many structures and functions.
5. Development is of many aspects (eg. physical, emotional, intellectual, social and moral) and individual differences are seen in the different aspects of development.
6. Rate of development is not uniform throughout the life span of an individual. Actually the child makes advancement during a particular period of development and takes rest in the next following period, to consolidate his development. In advancing therefore, he turns back and then moves forward again like a spiral. Thus it could be said that development is **spiral** and not linear.
7. Development is influenced by maturation and learning.
8. Development is very much related to one's environment. Stimulating environment helps an individual to develop many skills.
9. Development is based on modelling the parents, teachers and other elder members.
10. Since development is qualitative in nature, it cannot be measured directly. It could be assessed through keen observation in behavioural situations.
11. Development is possible even without growth. eg. Even short statured people may exhibit strong physical functioning.

1:10 Comparison between 'Growth' and 'Development'

Growth	Development
1) Growth is quantitative i.e. it indicates increase, enlargement and augmentation.	Development is both quantitative and qualitative and it results in improved functioning of the individual.
2) Growth could be objectively observed and measured.	As development is mainly qualitative, it cannot be measured; it could be assessed through keen observation in behavioural situations.
3) Growth does not continue throughout the life-span. It stops after maturation is obtained.	Development is continuous in all areas of mental activity; it does not stop after puberty.
4) Growth is one aspect of development.	Development is complex and many-sided.
5) Growth occurs in different parts of the organism.	Development describes the changes in the organism as a whole and does not list the changes in parts.
6) Growth is not directional.	Development is progressive and sequential.
7) Rate of growth is not uniform in all parts of the organism.	Rate of development is also not uniform.
8) Individual differences exist in growth among children.	Children differ in their level of development in different aspects-physical, social, emotional and intellectual.
9) Growth is not affected very much by learning.	Learning and experience affect development to a very great extent.
10) Growth may or may not lead to development.	Development in one aspect promotes development in other aspects (among physical, mental, social, emotional and moral) i.e. development is integrative.

1:11 Meaning of Maturation

Maturation is a natural process. It is the growth which takes place within the individual. The maturational changes are the result of unfolding and ripening of inherited traits and are relatively independent of activity, practice or experience. Maturation involves changes that are associated with normal growth.

1:12 Relationship between Growth, Development and Maturation

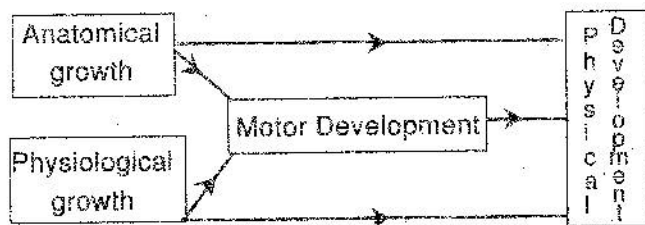
Growth refers to increase in size, length, height and weight of an organism. Growth is quantitative in nature and hence can be objectively observed and measured. Growth is one of the components of the developmental process. 'Development' implies overall changes in shape, form or structure, resulting in improved working or functioning of the organism. Development is progressive; series of changes occur in an orderly pattern. As growth in an organism continues, it reaches a maximum limit and stops after that. This **maximum level of growth** is referred to as 'Maturation'. Development is a continuous process, lasting throughout the life of the organism; it takes place both before and after maturation. Growth and maturation are largely determined by heredity. On the other hand environmental influences play a significant role in determining development.

In short, growth and development go hand in hand; but growth stops after maturation whereas development continues as long as the organism is alive.

1:13 Dimensions of Human Development

We can distinguish certain aspects of human development such as physical and physiological (bodily) development, with which are related locomotor and skill developments, cognitive (intellectual) development, emotional development and social development. These cut across and continually proceed through the various stages of development such as infancy, childhood, adolescence and adulthood. Major aspects of human growth and development are :

i) **Physical Development : Anatomical growth and Physiological growth** are the two components of Physical Development. The growth in the skeleton, changes in the quality and number of bones, changes in height, weight etc. constitute Anatomical growth. But the changes occurring in the internal organs like heart, lungs, liver, bladder, kidney, intestine, stomach, brain and groups of muscles are referred by Physiological growth. **Motor Development** refers to the growth and development of bodily organs like the limbs and the related muscles that activate them so that functional efficiency (**Manipulative skills**) is attained. Thus physical development consists of both physical and motor development. Even though there are subtle differences between physical development and motor development, they are generally referred to as Physical development as these two develop jointly and cannot be viewed in isolation.



ii) Intellectual or mental development : It includes the development of intellectual (also known as cognitive) powers like thinking and reasoning, imagination, concentration, creativity, sensation, perception, memory, association, discrimination and generalisation, etc.

iii) Emotional development : Under this aspect, starting with the basic instinct or emotion of '**General Excitement**', differentiation takes place during the evolution of primary emotions viz. joy, fear, and anger. Complex and secondary emotions like disgust, jealousy, elation, affection, anxiety, sympathy, distress, love, wonder, awe etc. are learned as the child grows and as opportunities occur. The emotional behaviour is developed to the point of emotional maturity and balance of emotions wherein the child learns to control emotion and express the proper emotion at appropriate situations. Use of language is also resorted to.

iv) Moral or character development : It includes the evolution of moral sense to discriminate the right and wrong and development of character. The individual develops his ethical and moral codes. Social development and moral development are intertwined.

v) Social development : Initially the child is self-centred and animalistic in character. Gradually he is developed into a social being, by learning to behave according to the rules and norms of society in which he lives. Due to social development he is able to get along with others in the society and participate in different social activities. He strives for social maturity.

vi) Language Development in Children : Language development in children appears to be dramatic. It does, of course, depend upon physical maturation, like control of mouth and tongue. Language development passes through certain stages though it is rather difficult to distinguish them clearly. The rate and time taken for language development exhibits individual differences. There appears to be specific sequence in language development. Children use basically the same rules of grammar during the early stages of learning any language (**Brown's study**), and so language abilities appear to mature just as motor abilities do.

The child speaks its first word normally at about one year, but even prior to this, it indicates its requirements by making various audible sounds and gestures. These form the **pre-lingual stage**, and include **cooing** and **crying**, babbling and gestures like smiling, moving head from side to side, etc. As the child learns to speak, gestures become less and less, but even later they are used to emphasize words.

By about 18 months, the child, on an average, uses 10 to 12 simple words. By the next 3 or 4 months, its

vocabulary increases rapidly and by 3 years, there is a sizable vocabulary in the mother tongue, which about 12 years increases to 10,000 words or more. Again, by 2 to 3 years, the child uses words to form sentences, uses pronouns and plural as well as speaking in past tense. By 4 to 5 years, sentence formation is considerably improved; simple rules of grammar are observed. After 5+, the school, along with the home, plays a vital role in language development, contributing to increased vocabulary, growth in sentence length, greater comprehension in reading, correctness in pronunciation, and in developing writing skill. Piaget has pointed out that in children we can notice a **shift from ego-centric to socialized speech** as they grow.

1:14 Development Phases

Of all the mammals, man is the most immature at birth and needs the longest period of development before he is capable of the activities and skills of his species. This prolonged period of immaturity points to the necessity of education. The human individual is a developing organism and progressive sequential changes take place in an orderly pattern from conception till death. These developmental changes could be best understood if we study them in different phases of human life, otherwise known as '**developmental stages**'.

1:15 Important Phases of Development

Based on certain common developmental characteristics psychologists have classified human life-span into the following developmental stages :

I Prenatal stage (from conception to birth)	[	Germinal Period (First 2 weeks)
		Embryonic Stage (III to VIII week)
		Foetal Stage (IX week to birth)

During the germinal period, the Zygote grows to $\frac{1}{4}$ ". Rudiments of bodily organs are formed during the embryonic stage and heart beat begins by about 8 weeks. During the Foetal stage, the child attains gradually its full size and shape. Cell division and differentiation characterise pre-natal development.

II Postnatal Stages

1. Infancy	From birth to 2 years
2. Childhood	Early childhood (From 3 to 6 years)
	Later childhood (From 7 to 10 years)
3. Pre-Adolescence	(From 11 to 13 years)
4. Adolescence	(From 14 to 17 years; in the strict sense, upto the onset of puberty)
5. Adulthood	Early Adulthood (From 20 to 40 years)
6. Middle-age	(From 40 to 60 years)
7. Old-age	(From 60 years onwards)

Postnatal development involves increase in size and diversity and complexity of functions. There are certainly individual differences and so we should not think that every child will have each stage according to the age mentioned.

The above 10 stages put forth by **Hurlock**, may be telescoped into 4 wider categories - Infancy (From birth upto 5 years), Childhood (5 to 12 years), Adolescence (12 to 18/20) and Maturity or Adulthood (after 18/20), as teachers may generally have to deal only with these broad stages of development in their professional work. It will also be of value to understand the general characteristics of individuals particularly during infancy, childhood and adolescence. These three stages are related to primary, middle and high school levels in education.

1:16 Physical Development from Infancy to Adolescence

The bodily growth is physical development. Physical development along with **physiological development** appears to be the basic aspect of human development. Physical development is rapid during infancy. A new born infant is about 50 cms in length and 3.5 kg in weight. It grows another 25 cms by the first year and weight becomes doubled during the first six months and trebled in the next six months.

Proportion of head to body changes from $\frac{1}{4}$ to $\frac{1}{8}$ and the lengths of arms and legs increase as the infant begins using them more and more.

Brain growth is more rapid. The brain has its full complement of cells even before birth and achieves 80% of its adult weight by 4 to 6 years. Another 13% is attained by the age of 8 with complete development at the end of adolescence.

The internal organs such as digestive organs, lungs, etc. grow rapidly and reach a balance of effective coordinated functioning. Sensory development goes along with such bodily growth and hearing appears well developed at infancy, though coordinated binocular vision takes more time. In caring for the infant's bodily development, mother's milk as also the affectionate atmosphere in which it is given and later nutritious food, cleanliness, adequate sleep, optimum sensory stimulation and bodily activities, are all vital.

The physical development secured during the first three years is strengthened and made more stable during early childhood. During later childhood bodily development appears to slow down a bit; body proportion approximates that of the adult. The stage of adolescence is again marked by an accelerated rate of physical growth along with the onset of sexual maturity. This growth is often uneven, with different rates resulting in clumsiness, awkwardness in movements and self-consciousness. Girls and boys differ regarding physical growth. Man reaches a greater average level in height and weight than woman although at the age of 13 girls surpass boys in average weight since girls attain puberty about a year or two earlier than boys do.

Motor activities such as head control, movement of eyes, sitting, standing, walking, running, climbing (all these are referred as **locomotor skills**) and **manipulative skills** like 'handling of objects with dexterity', 'eye-hand co-ordination', precision of control,

speed of arm movement, arm steadiness, aiming, etc. are important for personal and educational growth. The child makes contact with the external world only through motor development which is closely related to intellectual, emotional and social development:

There are individual differences in the growth and development. An early discovery of the problems and deviations in physical development can help the teachers to give proper guidance in promoting adjustments among children. Proper physical and motor development promote self-help and competence in children.

1:17 Intellectual Development from Infancy to Adolescence

Intellectual development is also referred to as cognitive development. Intellectual development refers to the gradual growth in what are called cognitive abilities (ability to attend, to perceive, to discover, to recognise, to imagine, to judge, to conceptualise, to remember, to learn, to reason, to take decisions, to solve problems and to indulge in meaningful speech) and also to consequent growth in knowledge and adjustment to environment. Cognitive development is influenced by nutritional, emotional and social factors.

The mental powers of the new born infant are immature and undeveloped; its brain centres are still maturing and its knowledge at birth is practically nil. But even early, observers can notice individual differences in the mental powers of infants due to hereditary differences.

The infant's first intellectual response is to stimuli from the outside world. Its intellectual activities broaden as its power of imagery grow and powers of retention of past sensory experiences and their recall develop. Imagination which comes to the forefront by about the age of three, expresses itself through play activities and it also serves as a means of intellectual development. Language (mother tongue) helps in preserving as well as fostering intellectual growth which continues to accelerate as children develop the ability to judge, to reason and to solve problems, starting from the concrete level and going towards the abstract. Creativity helps in divergent thinking, leading to novel intellectual products and practices. Psychologists like **Jean Piaget** point out that **child's cognitive processes are not abridged or simplified versions of adult processes but are qualitatively different**, though certain common invariants are also there. Broadly speaking, cognitive development of children can be fostered by a stimulating home and school environment with a lot of scope for free self activity and opportunities for varied sensory experiences.

1:18 Social Development from Infancy to Adolescence

Social development takes place gradually. Being a social person is not innate; it must be learned. The new born infant exhibits little or no social activity. It is concerned with its satisfaction of bodily needs. At birth it is non-gregarious and the '**I-feeling**' alone appears to be dominant. Even about 4 weeks, the infant stares at

faces closely and a few weeks later wants to be picked up. The first significant attachment behaviour of the infant is recognition of its mother and gradually this social interaction circle widens to include other members of the family and the '**we-feeling**' emerges.

The early play activities of very young children is solitary play; later on it becomes parallel play (each child in a group playing in its own way). Only by about the age of two, the child shows a desire to cooperate with others in group plays. Thus early childhood is **pre-gang stage** of social development but childhood is the gang stage, when playmates have an active role in accelerating social development. No sex differences is felt at this stage. The child is assimilating socializing experiences through direct imitation of the patterns of other children and by spontaneous reaction to things which they do or to situations they create. Social behaviours that are developed during childhood are rivalry, aggression, co-operation, sympathy, social approval and certain social prejudices imbibed from adults. Notions of proper and improper social behaviour gradually develop.

Later childhood (8-12) is marked by maximum social adjustment in the changeable and variable social world marking a level of **pseudo maturity**. Though the child stays away from its home now a considerable amount of time, the importance of paternal affection continues to be undiminished.

Adolescence is marked by sex consciousness in social relations. Adolescents of the different sexes try to

mutually attract each other. But relations between boys and girls are not very well defined during early adolescences resulting in a lot of embarrassment in the behaviour of youth. Popularity or unpopularity play a vital part in adolescent adjustment. Self-assertiveness, social insight, social competence, social discrimination or snobbishness are all seen emerging during later adolescence and ultimately adult social patterns develop. Adolescent gangs are also powerful factors in socialisation and peer group culture is dominant. This may sometimes lead to antisocial or delinquent behaviour, if not guided and properly sublimated.

Developmental psychologists have listed certain social characteristics of children at different stages of development. Pre-school children have one or two friends but change their friends easily as they are socially flexible. Play groups at this stage are small and loosely organised. Quarrels are frequent but are of short duration and quickly forgotten. Dramatic play is frequent and awareness of sex roles begins at this stage.

Primary class children appear to be more selective in their choice of friends and like organised games; quarrels are still frequent and words are more often used than physical aggression. Competition and boasting are often seen and boys and girls show different interests, both in work and play.

During elementary grades, the gang influence becomes powerful. Interests of boys and girls become more divergent. Team games are more popular.

During adolescence, peer group dominance becomes more pronounced and concern for opposite sex increases. At the end of adolescence, adult social patterns come out.

1:19 Emotional Development from Infancy to Adolescence

The new born infant has a single generalised emotional response of excitement only and out of this gradually evolves increasing differentiated patterns of emotional expression over the course of development. In about 6 months, the child can express the three primary emotions as described by **Watson** :

- thwarting of appetites and restricting the free bodily movements of the infant naturally makes it **angry**;
- loud, harsh sound and sudden loss of support evokes **fear**;
- Satisfaction of appetites, petting and fondling produces love and **joy**.

Social factors, imitation and learning lead to a variety of emotional reactions and expressions and by early childhood most of the common emotions-affection, elation, joy, anger, disgust, jealousy, fear, sympathy, etc. are clearly discernible.

Pre-school children, because of their immaturity and lack of control, tend to express their emotions freely and openly. Angry outbursts, which are seen frequently, are

caused by fatigue, hunger or needless adult interference. As they are often exposed to many strange situations, they exhibit many fears and their imagination produces a variety of irrational fears also in such children. Jealousy is equally common at this stage. Primary grade children (between 6 to 9 years) are often alert to the feelings of others; but they often hurt the feelings of other children. They are very sensitive to criticism and need praise and recognition. After the age of 9, conflict between group codes of the peers and the adult rules often cause emotional difficulties.

At the end of infancy, changes occur in the mode of expression of emotional reaction. For example small infants express their emotion of anger through crying. At the age 3, on becoming angry, infants start beating or biting others. At the age of 4, they show their anger by insubordination. At the age of 6, they vent their anger through verbal abuses. Such patterns of change in emotional reactions exhibited by children could be noticed in other emotions also.

According to Freud there seems to be a twin faced emotional behaviour of individuals during infancy and early childhood. Male children display much affection towards their mother. Though they have affection towards their father, they also have jealousy and hatred towards him, as they feel father has more concessions with the mother, whom they love most (This is known as **Oedipus complex**) Similarly in girls appear the opposing emotions of love and hatred simultaneously towards their mother.

(This is known as **Electra complex**). If such emotional complexes are intense and persist for long, they may affect the personality characteristics of the individuals later on.

1:20 Stages of Moral Development in Children

Psychologists mention 4 stages in the moral development in children. In the first stage, known as '**Prudential Stage**' children's behaviour is regulated by means of pleasure and pain. Children learn those behaviours that are rewarded and avoid those which are punished. The second stage is the '**Authoritarian Stage**' in which children follow the directions of his elders like parents and teachers who have dominating influence over the children. The third one is the '**Social stage**' in which children's behaviour is governed by the social acceptance or rejection. Children try to behave in conformity with social expectations. The level of moral development in most of us get stagnated at this stage. The highest stage of moral development is the '**Personal Stage**' in which an individual's behaviour is influenced by his conviction of high morals and ethics. Even if the whole society opposes him he will be steadfast in his action, if he thinks his action is highly ethical i.e. he will be courageous in his convictions.

1:20:1 Jean Piaget's Stages

Jean Piaget and **Laurence Kohlberg** have put forward a development theory of morality. Piaget distinguishes between '**moral realism**' and '**moral**

relativism'. Young children are moral realists who interpret moral rules and look at the material side of the actions and their magnitude, ignoring the intentions of the acts. Suppose a child is asked to pass moral judgements on two acts. In the first case a child breaks two tea cups while attempting to reach stealthily the sweets kept over the cup board and in the second case a servant maid breaks 4 tea cups due to carelessness while serving tea to the visiting guests. The child will hold the servant maid as more guilty than the stealing child because of the magnitude of the act i.e. breaking more number of cups.

Only by about 11-12 years, do children become capable of moral relativism. They are now capable to decentre and think of several qualities at once and they have overcome their **ego centricism** by becoming aware that other people may have different points of view. Consequently they take into account motives and circumstances in making moral judgements. After the age of 12 children behave in a conciliatory and forgiving way.

1:20:2 Kohlberg's Stages

Kohlberg's theory of moral development is based on Piaget's general theory of cognitive development (Kohlberg believes that moral development, like cognitive growth takes place through 'assimilation' and 'accommodation') He lay emphasis on the significance of looking at things from other's point of view (role taking). There are three levels of moral development.

In level 1, the **preconventional level**, Kohlberg regards it is based on the desire to avoid punishment and gain rewards. At level 1 there is no internalization of moral standards. At level 2, the **conventional level**, although the child identifies with his parents and conforms to what they regard as right and wrong, it is the motive to conform rather than ethical standards that have been internalized. It is only at level 3, the **postconventional level**, that moral judgement is rational and internalized and that conduct is controlled by an internalized ethical code and is relatively independent of the approval or castigation of others. At this level, moral conflict is resolved in terms of broad ethical principles, and violating these principles results in guilt and self-condemnation.

Like Piaget, Kohlberg also speaks of maturation as the basis of moral growth; but mere maturation may not help one to reach the higher levels of morality. The individual must undergo life experiences that specifically "confront him with more complex moral dilemmas and with the realities of the social system in which he lives". These points are of relevance in planning a course of moral education. Kohlberg advocates the existence of some basic or universal moral values as all normal children everywhere go through the same stages of cognitive development and interact socially in essentially the same ways.

1:21 Period of Adolescence

The word 'adolescence' comes from the Latin verb

'*adolescere*' which means 'to grow'. So the essence of the word adolescence is growth and it is in this sense that adolescence represents a period of intensive growth and change, in nearly all aspects of child's physical, mental, social and emotional life.

Adolescence begins with the onset of puberty and lasts till the beginning of adulthood (12+ to 18/20). Several physiological and psychological changes occur during this period. The spurt in growth during adolescence starts with the secretion of hormones from the ovaries, testes and adrenal glands into the blood stream and tissues. According to Cole, adolescence is a period of growth in all systems of the body. In the course of a few years the individual undergoes changes both in size and in his internal body chemistry. The rapidity, variety and force of these developments are amazing. The changes are so extensive that some people call this as second birth. Stanley Hall called this period as "**period of storm and stress**", a time when the individual is erratic, unstable and unpredictable. But modern researches and interpretations have made much of Hall's views obsolete. Erickson's epigenetic model and self-concept development have added new dimensions to the interpretation of adolescence.

1:22 Special Characteristics of Adolescents

1) **Rapid perplexing physical changes** : Rapid physical changes in physical appearance, development of internal organs, functioning of sex organs facilitating

the individual to acquire parent hood, physical strength reaching its zenith etc. take place during adolescence.

ii) Heightend Self-consciousness : Adolescents are highly anxious about the physical changes that take place in them and try to present themselves most attractive in all possible ways (hence referred to as an age of **self-decoration**). They attempt to attract others through their dresses, make-up, manner of talking, walking etc.

iii) Intensification of sex-urge : Sex consciousness becomes too intense at this time. Most of the adolescents' problems are concerned with the sudden functioning of their glands, secretion of sex hormones and the awakening of the strong sex instinct.

iv) Craving for Independence : The adolescent is on the boundary line demarcating childhood from adulthood. He is typically a person who needs security, guidance and support from parents like a child and at the same time he has independent views of his own; wishes to have economic self-dependence and autonomy in his functioning. He starts reacting strongly when parents and elders treat him as a child. He tries to assert himself and displays dissent to being controlled by powerful others. He possesses a strange mixture of dependence and independence which creates conflicting situations and problems for him.

v) Peer - group relationship : Peer group relationship plays a substantial role in the life of an

adolescent. He gets distanced from his parents and elders and spends much of his time with the members of his peer group. He values the ideals of the group and develops a sense of loyalty towards it; he is now directed by the standard and norms of his peer group. When there is a conflict between the demands of social and cultural norms of adults and those of the peer groups, he gets frustrated, which may at times lead to his maladjusted behaviour. The group behaviour of adolescents in schools/colleges is on any one of the two lines:

- a) fun and joy through activities like singing, dancing, playing, seeing movies, etc.
- b) indulging in destructive activities like noisy protests, strike, teasing others, etc.

vi) Idealism vs. Realism : A typical feature of adolescents lies in their interest in ideals. But in this search of idealism they go away from realism. Lack of experience makes him somewhat unrealistic. He tends to accept the impossible. (eg. Insisting that everyone should come in the queue to receive their ration is laudable; but denying exemption for the sick and the aged from this rule reflects one's unrealistic attitude). When it is not attainable he becomes disturbed and unreasonable.

vii) Vocational Choice and Need of Self-support : The period of adolescence requires from the individuals to decide about their vocations. Emotional instability, lack of experience and maturity prove to be obstacles in making the right choice. Moreover their interests,

aptitudes and abilities, make them puzzled with respect to exercising their vocational choice. In the absence of proper guidance they may turn out to be misguided youth indulging in anti-social activities.

viii) Aspiring for Leadership : Adolescents are stimulated by idealism and aspire for democratic leadership. They are unselfish, responsible and respect other's points of view. These qualities of leadership are present in them in a natural way which could be further developed by entrusting to them the responsibility for organising various team events like games, excursion, debates, cultural competitions etc. within the school as well as between schools, under the guidance of teachers.

1:23 Problems of Adolescents

Adolescence is a period of stress and strain. All are not able to smoothly cross over this stormy period of life. Important problems of adolescence are discussed below :

1:23:1 Identity Crisis

Though an adolescent needs security, guidance and economic support from his parents, still he tries to establish to others that he has independent views, maturity of opinion and relies on self-support like adult. Thus his conflicting stances lead him to self-confusion about his own identity (whether he is a child or an adult). During later adolescence, he slowly tries to come out of this confusion about his self identity and develops his own norms of behaviour and expectations; develops his abilities and values.

1:23:2 Generation Gap

In today's changed world, there is a large gap between the view points of the elders and youngsters of this generation. Progress in science and technology and the consequent increase in the life comforts of our adolescents have brought in a sea change in the goals, actions, ideals, interests, values and expectations of our youngsters which has alienated them from the present day elders. This alienation is termed as generation gap which psychologists consider as responsible for the conflicts between the adolescents and their elders at home as well as in the society at large [(eg.) youngsters want to pay respect to others, not on the basis of age or seniority but due to efficiency, ability and performance which is not agreeable to elders].

Rapid technological changes witnessed in our society coupled with the craze for independence have unfortunately contributed a lot to the denigration of our age old customs, cultural traditions and values in the midst of youngsters. Many a social institution like family and school has undergone changes in their structure and functioning, resulting in increased opportunities for our adolescents to go astray and indulge in smoking, drinking, drugging, developing liberal sexual relations etc.

1:23:3 Negative and Non-Conformist Attitude

In their attempt to assert their individuality, adolescents, oppose all the elders, particularly those who control them. They question every social tradition and do

not meekly conform to all social customs, which create problems in their social adjustment.

1:23:4 Domination of Sexual Urge

The foremost problem in adolescence is the excitement created due to the domination of sexual urge, which leads to distraction and wastage of energy of the youths.

1:23:5 Hesitation and Confusion in Selecting a Vocation

As they are not sure about their own potentials, interests, abilities and aptitudes and also proper guidance is not available in knowing about different occupations, their nature and scope, requisite entry qualifications and skills etc. adolescents appear to be highly confused in their selection and preparation for future career.

1:23:6 Emotional Swings

Adolescents are highly restless, emotionally perturbed and moody; without any proper reason, at one time they appear to be elated and joyous and the next moment highly depressed. Unfulfilled desires and longings make them aggressive. As they aspire for those beyond their abilities and potentials, easily they get frustrated. Adolescence is said to be a stage of "ignorance of one's practical world".

1:23:7 Aspiring for Autonomy

Adolescents ardently wish to be liberated from parental control and function independently. They feel

they are grown up and mature and also expect due recognition from others. They display dissent to being controlled or supervised. The greatest social problem of adolescents is that they start questioning their parents, become disobedient and protest strongly when being controlled. In severe cases, they stay away from their family or run away from the home.

1:23:8 Peer Group Dominance

Adolescents get distanced from their parents and elders and spend much of their time with the agemates of their peer group. They value the ideals of the peer group and develop a sense of loyalty towards it. Their behaviour is directed by the standard and norms of the peer group. In the company of bad friends, adolescents derail in their social behaviour, at times leading to delinquency. Hence parents are required to help their adolescent children in selecting their friends and channelise the energy of the peer group in constructive and creative activities.

1:24 Helping the Adolescents to Overcome their Problems

The needs of the adolescents have to be satisfied and their problems are to be realised in a proper way in order to help them in growing and developing properly. This difficult task requires the joint efforts of parents and teachers. Below are some suggestions in this regard:

i) To have the proper knowledge of adolescent's psychology : Adolescence is the bridge between childhood and adulthood. It is essential to have the knowledge of the adolescent's psychology in order to understand them and their problems. They should be treated sympathetically and helped to solve their problems by offering proper guidance. They should be helped in resolving their conflicts and overcoming frustrations.

ii) Providing suitable environment for proper growth : Adolescence is the period when maximum growth and development takes place. To optimize this development, suitable environment should be provided by the parents and teachers at home as well as in schools. Adolescents must be provided with balanced diet; they must be given knowledge of health and personal hygiene, cleanliness, various diseases and their prevention etc. to keep them fit and agile. Adequate provision for physical exercise, games and recreations should be made in the school curriculum and facilities made available. Making use of their interests, curricular and co-curricular experiences are to be planned and arranged.

iii) Rendering proper sex education : Sex plays dominant role during adolescence. As adolescents are frightened by nocturnal emission/menstruation, they should be provided with proper sex information and education; or otherwise they may become victims of exploitation by yellow journals, **pseudo sexologists and advertising agencies which provide half-backed information on sex.**

iv) Proper dealing with adolescents :

- a) Parents and teachers should recognize the importance of the peer group on modifying adolescents' behaviour.
- b) Elders should not attempt to thrust their views and decisions on the adolescents; instead they should try to convince them by their rational arguments.
- c) As adolescents crave for recognition, teachers and parents should invite their views and ideas and give due consideration and recognition.
- d) As adolescents wish for social recognition, rights and responsibilities, opportunities are to be provided for them to carry out team work, undertake field trips, participate in N.C.C. Camps; fine arts etc; organise school celebrations and exhibitions and so on.
- e) Teachers and parents should serve themselves as role models for adolescents in the religious and moral spheres, instead of criticising / reprimanding them.

v) Training of emotions and satisfaction of emotional needs : Adolescent youths are surcharged with emotions, highly inflammable and restless. Their emotions can be aroused with a slight provocation. Political parties and opportunist leaders can easily befool them and instigate them to indulge in strike and damage

public properties. Therefore there is a strong need for proper emotional education. Their emotions should be properly trained and their emotional energies should be diverted towards constructive ends. Provision in schools for students' self-government, fine arts association, radio-club, science club, science exhibition etc. provide opportunities for the youths to develop their organising and leadership skills. Similarly participation in scouts, N.C.C. social and community service, sports and games, competitions, quiz etc. offer opportunities to develop their abilities and interests.

vi) To take care of the special interests of the adolescents : A great care should be taken to locate the special interests and aptitudes of adolescents. According to their interests and aptitudes they should be provided with learning experiences and opportunities for participation in co-curricular activities.

Their thirst for curiosity, wandering and adventure should be quenched through excursion, N.C.C, mountaineering, scientific exploration etc. Their love for humanity and ideals should be utilised in rendering social and community services like blood donation camps, functional literary movements etc. in the neighbourhood and distressed areas so as to make them constantly busy and their mind pre-occupied with healthy and constructive ideas.

vii) Providing Religious and Moral Education : Opportunities should be provided for adolescents to

discuss religious and moral issues rationally. Emphasis should be shifted from rituals to ideals, by facilitating a comparative study of the tenets of some important world religions. By nature, adolescents are hero worshippers. If we provide them with selected examples of saints and prophets, religious and social reformers and great leaders who sacrificed their lives in our struggle for independence, their interests can be better served as they will imbibe many of the ideals of their heroes.

viii) Provision for Vocational Education : As adolescents aspire for economic self-dependence, they should be provided with proper vocational guidance to take up jobs during their holidays and vacations. Vocational oriented curriculum at the secondary level will be of great help in grooming the adolescents.

ix) Arranging Guidance Service : Adolescents are at the cross-roads of life. Lack of guidance creates feelings of insecurity and restlessness in them. Hence in all high schools and colleges, guidance cells should be established and made functional, offering educational, vocational and personal guidances to our students and youths.

1:25 Developmental Tasks

Every culture expects its members to master certain essential skills and acquire certain appropriate behaviour patterns at various ages during the life span. These tasks, skills and behaviour pattern are essential to personal and social adjustment at those ages and each cultural group

expects its members to acquire these. For example all children are expected to master the 3 R's (reading, writing and counting) during their early childhood years. Heavinghurst has labelled these as "**Developmental tasks**". **Developmental task** is "a task which arises at a period in the life of an individual, successful achievement of which leads to his happiness and success with later tasks, while failure leads to unhappiness and difficulty with later tasks". Developmental tasks are different for different developmental periods of an individual namely, childhood, adolescence, adulthood, middle age and old age. Some developmental tasks arise purely from maturation (eg. learning to walk), others arise from socio-economic pressures (eg. learning to read) and a few may grow out of personal values and aspirations of the individual (eg. choosing some vocation). In most cases developmental tasks arise from these forces working together. Schools should help students to deal with the tasks appropriate to their level of development.

The ages at which the social group expects its members to master the developmental tasks of that age are called **critical ages**. During such critical periods, maturational factors predispose the individual to acquire these patterns of behaviour which facilitate adjustment to the society and external world. As values undergo changes in society, some of the old accepted developmental tasks get eliminated and may be replaced by new tasks or their relative importance may get lessened.

1:25:1 Uses of the Knowledge of Developmental Tasks

- i) Developmental tasks are of value to the individual in enabling him to know what is expected of him by society at a particular age.
- ii) Parents who are aware of such developmental tasks may teach them to their children so that the children may not experience failure and unhappiness later.
- iii) The developmental tasks show the individual what lies ahead and what he will be expected to do when he reaches the next stage of development and this will eliminate much of the difficulty and emotional stress in later adjustment. For example, if adolescents are aware of the problems and tasks that lie ahead, they could be better prepared for them.

If parents and teachers know the developmental tasks, then they can provide the necessary experiences and training to children and ensure the children to behave as per the expectations of the society and lead a successful life. If any child is found to lag behind in developmental tasks at a particular period, then it can be provided with special education or training. For example, we could often hear in our countryside old ladies and grand mothers enquiring their relatives, "Has your child completed two years of age? Is he talking now?" etc. This is done precisely for the purpose of knowing the

developmental trend of the child. If a two year old child is not talking, then it points out to the necessity of referring the child to a doctor and procure the help of a speech therapist.

1:25:2 Harms due to Insistence of Developmental Tasks

- i) Inappropriate expectations : Developmental norms may not hold good in the case of late or slow developing children like the retarded. Further, showing disappointment when a particular behaviour pattern does not show itself in a child at the normally expected time may result in feelings of inferiority and consequent further retardation in the child.
- ii) Among the developmental tasks, omitting a few earlier one's and attempting to develop the next higher tasks in children may prove to be injurious for the proper development of the child. For example, in some schools in their over enthusiasm, in the middle of XI Standard without finishing the entire syllabus meant for that class, they start teaching the syllabus of XII Standard. This is highly injurious to the proper intellectual development of children.
- iii) When an individual passes from one developmental stage to the next higher one, his unique experiences and the resulting frustration may sometimes prove to be harmful.

For example if a 25 year old lady does not agree to marry or determines to remain unmarried, then our society talks ill of her, considers her suffering from defects or illness. This is because the developmental norm in our society for a 25 year old girl is to get married. This kind of criticism may lead to conflicts and frustrations in the lady and her life may be shattered.

1:26 Developmental Tasks for the Various Stages of Development

Havinghurst has listed the following developmental tasks for the various stages of development. It has to be remembered that some of these may not be pertinent to our culture system and life values, as such tasks have a cultural bias.

- a) **Developmental tasks of infancy and early childhood :**
 - Learning to walk, to take solid food, to talk, to control the elimination of body wastes.
 - Learning sex differences and sexual modesty.
 - Achieving physiological stability.
 - Forming simple concepts of social and physical reality.
 - Learning to relate oneself emotionally to parents, siblings and others.
 - Learning to distinguish right from wrong and develop conscience.

b) Later childhood :

Learning physical skills necessary for ordinary games.
Building wholesome attitudes towards oneself as a growing organism.

Learning to get along with age-mates.

Learning appropriate sex roles.

Developing fundamental skills in reading, writing and calculating.

Developing concepts necessary for everyday living.

Developing attitude towards social groups and institutions.

c) Adolescence :

Accepting one's physique and accepting a masculine or feminine role.

Establishing new relations with age-mates of both sexes.

Gaining emotional independence from parents and other adults.

Achieving assurance of economic independence.

Selecting and preparing for an occupation.

Developing intellectual skills and concepts necessary for civil competence.

Desiring and achieving socially responsible behaviour.

Preparing for marriage and family life.

d) Adulthood :

Selecting the life partner.

Entering into the family life.

Begetting children and rearing them up.

Getting in to an occupation or profession.

e) Middle Age :

Shouldering the family and social responsibilities.

Acquiring economic stability in life.

Providing guidance to one's own adolescent children.

Getting adjusted with one's own old age parents.

f) Senescence (old age) :

Accepting decrease in one's own capabilities and efficiency.

Withstanding the loss of one's life partner.

Developing positive attitude towards leisure time and relaxation.

1:27 Education and Developmental Tasks

The primary aim of education is to provide experiences and training to children so as to equip them with necessary basic skills and knowledge, due to their age, thereby enabling them to lead a successful life. This gigantic task is undertaken at different stages of education viz. elementary, secondary, higher secondary and higher education levels. As the rate of learning of an individual is in accordance with his physical and

intellectual development, it is the duty of the school to arrange for adequate opportunities for the child and motivate him to acquire the requisite knowledge and basic skills at every stage of education. Educationists should take special care in designing the school curriculum for different classes ensuring it to be broadbased and capable of providing variety of experiences through curricular and co-curricular activities. The knowledge of developmental tasks enable the lead players in education viz. parents and teachers to understand the complexities involved in the teaching - learning process and develop that of an allround personality in the child. This make them realise the importance of children participating in sports and games, physical exercises and various co-curricular activities as well as the need for cultivating personal health and hygiene.

Conclusion

In this unit, Meaning and Definition of Psychology, Historical development of Psychology and its emergence as a separate discipline, Meaning and Definition of Growth and Development, their characteristics, Comparison between Growth and Development, Meaning of Maturation, Relationship between Growth, Development and Maturation, Dimensions of Human Development, Definition of Educational Psychology, Scope and Importance of Educational Psychology, important stages of human development, their characteristics and

developmental tasks at various stages of human development viz. infancy, childhood, adolescence, adulthood, middle age and old age, and education and developmental tasks have been discussed elaborately.

Review Questions :

- 1) Define the term 'Educational Psychology' and discuss the need for knowledge of educational psychology to classroom teachers. (A) [Ans. 1:03 + 1:07]
- 2) Explain the meaning of the term 'Psychology'. (C) [Ans. 1:01]
- 3) Explain briefly the birth of Psychology and its emergence as a separate discipline. (A) [Ans. 1:02]
- 4) Define Educational Psychology. (C) [Ans. 1:03]
- 5) Explain the nature of Educational Psychology. (C) [Ans. 1:04]
- 6) Explain the scope of Educational Psychology (A) [1:05, particularly those under 2]
- 7) What are the focal areas of Educational Psychology? (C) [Ans. In 1:05 those, under 2 only i.e. till the end of (v) The Teacher]
- 8) Define 'Growth' and 'Development' (C) [Ans. 1:08 + 1:09]

- 9) Explain the differences between Growth and Development (B) [Ans. 1:10]
- 10) Explain the characteristics of growth and development (B) [Ans. 1:08:1 + 1:09:1]
- 11) Explain the different areas or dimensions of human development (A) [Ans. 1:13]
- 12) Differentiate Physical development from Motor development. (C) [Ans. In 1:13 (i)]
- 13) Differentiate Anatomical growth from Physiological growth (C) [Ans. In 1:13 first 8 lines]
- 14) Explain Social Development (C) [Ans. In 1:13(v)]
- 15) Write short note on the following:
 - i) Mental Development (C) [Ans. 1:13 (ii)]
 - ii) Emotional Development (C) [Ans. 1:13 (iii)]
 - iii) Moral Development (C) [Ans. 1:13 (iv)]
 - iv) Language Development (C) [Ans. In 1:13 (vi)]
- 16) Explain the relationship between Growth, Development and Maturation. (B) [Ans.: 1:12]
17. Mention the important stages of physical development. (C) [Ans. 1:15]
18. Explain briefly the physical development of the individual from infancy to adolescence. (B) [Ans. 1:16]
19. Discuss the intellectual development of the individual from infancy to adolescence. (B) [Ans. 1:17]

20. Describe the social development of children from infancy to adolescence. (B) [Ans. 1:18]
21. Describe the emotional development of children from infancy to adolescence. (B) [Ans. 1:19]
22. Describe the stages of moral development in children. (B) [Ans. 1:20 + 1:20:1 + 1:20:2]
23. Explain the significance of adolescence in human development. (B) [Ans. 1:21]
24. Describe the special characteristics of adolescence. (B) [Ans. 1:22]
25. Discuss the important problems of adolescence and how parents and teachers could help the adolescents in overcoming their problems. (A) [Ans. 1:23 + 1:24]
26. Mention any four important problems of adolescents. (C) [Ans. 1:23 + 1:23:1 to the end of 1:23:4]
27. "Adolescence is a period of storm and stress". Discuss this statement in relation to their characteristics and the problems they face. (A) [Ans. 1:21 + 1:22 + 1:23]
28. What are developmental tasks? (C) [Ans. 1:25]